SRS — ms_04 Retail

1. Fungsi

Service inti sisi toko: **POS/penjualan**, **stok retail**, **permintaan restock** ke warehouse, penerimaan barang distribusi, serta **dashboard** (penjualan, laba-rugi, analisis barang) berbasis read model.

2. Kebutuhan

- Master retail (cabang) dan stok per barang dengan `min_stock`.
- Penjualan multi-item: validasi stok, snapshot `unit_price` dan `unit_cost` dari Commodity, reserve stok saat PENDING.
- Sale menjadi COMPLETED setelah event `payment.paid`; CANCELLED setelah `payment.failed` (reservasi dilepas).
- Saat stok ≤ `min_stock` menerbitkan `retail.stock.low`; stok habis ditolak saat penjualan.
- Manajer membuat `restock_requests`; menerima barang saat `shipment.delivered` (stok bertambah) lalu konfirmasi.
- Dashboard (peran OWNER: semua retail; RETAIL_MANAGER: retailnya):
  - penjualan harian/bulanan, jumlah transaksi, rata-rata nilai transaksi;
  - laba-rugi = pendapatan − HPP (`cogs`), keseluruhan dan per retail;
  - analisis barang: terlaris, lambat laku, kontribusi laba, per kategori.

3. Endpoint

Method	Path	Deskripsi	Peran
CRUD	`/retails`	Master retail	OWNER
GET	`/stocks`	Stok retail	CASHIER, RETAIL_MANAGER
POST	`/sales`	Buat penjualan	CASHIER
GET	`/sales`, `/sales/{id}`	Riwayat/detail	CASHIER, RETAIL_MANAGER, OWNER
POST	`/sales/{id}/cancel`	Batalkan sale PENDING	CASHIER
POST	`/restock-requests`	Ajukan restock	RETAIL_MANAGER
GET	`/restock-requests`	Daftar/status	RETAIL_MANAGER, OWNER
POST	`/restock-requests/{id}/confirm`	Konfirmasi penerimaan	RETAIL_MANAGER
GET	`/dashboard/overview`	Ringkasan penjualan & laba-rugi (`from`,`to`,`retail_id`)	OWNER, RETAIL_MANAGER
GET	`/dashboard/retails/{id}/profit-loss`	Laba-rugi per retail	OWNER, RETAIL_MANAGER

Method	Path	Deskripsi	Peran
GET	`/dashboard/retails/compare`	Perbandingan antar retail	OWNER
GET	`/dashboard/items/analysis`	Analisis barang terjual	OWNER, RETAIL_MANAGER

4. Model Data

`retails`, `retail_stocks`, `sales`, `sale_items`, `restock_requests`, `restock_request_items`, `sales_daily_summaries`, `item_sales_summaries` — lihat er.puml.

5. Event

- Publish: `sale.completed`, `retail.stock.low`, `restock.requested`.
- Consume: `payment.paid`, `payment.failed`, `shipment.created`, `shipment.delivered`, `sale.completed` (internal, pembaruan ringkasan dashboard).

6. Aturan Bisnis

- `profit = Σ (unit_price − unit_cost) × qty`; `cogs = Σ unit_cost × qty`.
- Pembaruan ringkasan idempoten (tabel `processed_events` berdasarkan `event_id`).
- Restock request hanya untuk retail milik manajer (cek `X-Retail-Id`).

7. Diagram

usecase.puml · er.puml · sequence_sale.puml · sequence_restock.puml · sequence_dashboard.puml · state_sale_restock.puml

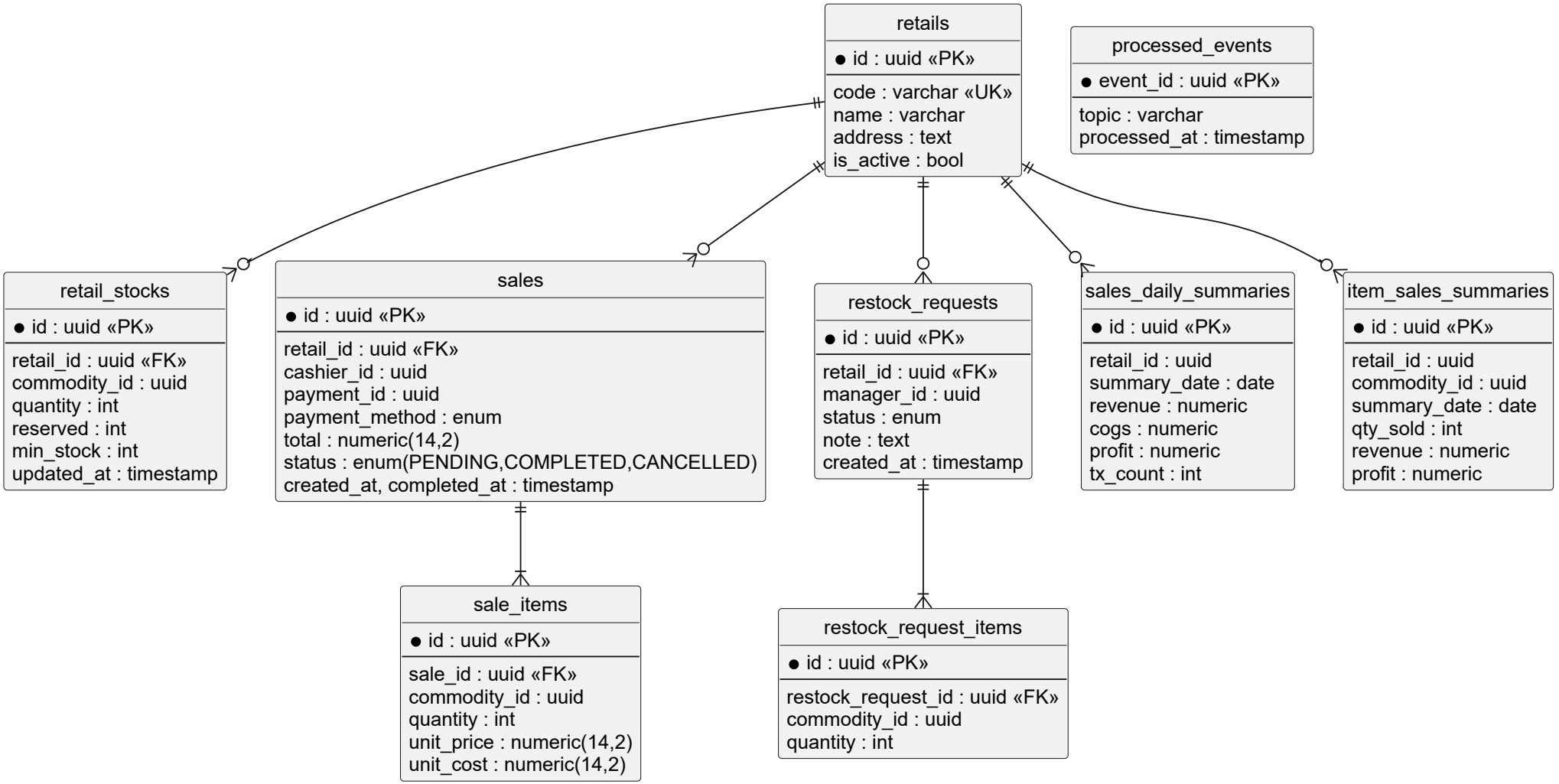
8. Cara Pembuatan

1. Database `db_retail`; migrasi seluruh tabel + `processed_events`, `outbox_events`; indeks `(retail_id, created_at)` pada `sales`, unik `(retail_id, summary_date)` pada ringkasan.
2. Modul stok: `Reserve`, `Release`, `Deduct`, `Add` dengan `SELECT … FOR UPDATE` agar tidak oversell.
3. Modul penjualan: client Commodity & Payment (timeout + retry), mesin status sale.
4. Consumer event + publisher (outbox).
5. Modul restock request.
6. Modul dashboard: handler `sale.completed` mengisi `sales_daily_summaries` dan `item_sales_summaries`; endpoint membaca tabel ringkasan, lalu memperkaya nama barang via Commodity.
7. Uji: concurrency stok, idempotensi event, perhitungan laba-rugi, otorisasi per retail.

Lampiran Gambar Diagram

ER - Retail

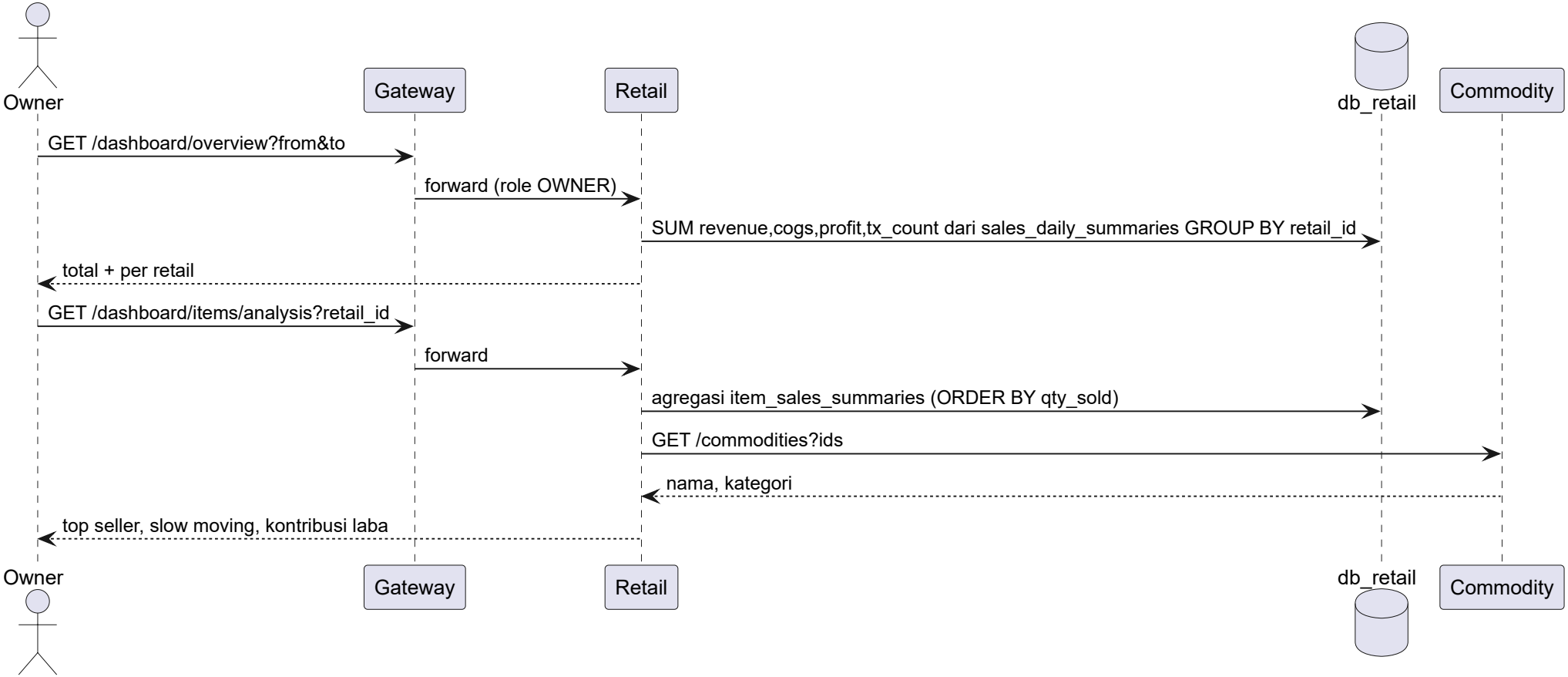
ER - Retail


retails
● id : uuid «PK»
code : varchar «UK»
name : varchar
address : text
is_active : bool
processed_events
● event_id : uuid «PK»
topic : varchar
processed_at : timestamp
retail_stocks
● id : uuid «PK»
retail_id : uuid «FK»
commodity_id : uuid
quantity : int
reserved : int
min_stock : int
updated_at : timestamp
sales
● id : uuid «PK»
retail_id : uuid «FK»
cashier_id : uuid
payment_id : uuid
payment_method : enum
total : numeric(14,2)
status : enum(PENDING,COMPLETED,CANCELLED)
created_at, completed_at : timestamp
restock_requests
● id : uuid «PK»
retail_id : uuid «FK»
manager_id : uuid
status : enum
note : text
created_at : timestamp
sales_daily_summaries
● id : uuid «PK»
retail_id : uuid
summary_date : date
revenue : numeric
cogs : numeric
profit : numeric
tx_count : int
item_sales_summaries
● id : uuid «PK»
retail_id : uuid
commodity_id : uuid
summary_date : date
qty_sold : int
revenue : numeric
profit : numeric
sale_items
● id : uuid «PK»
sale_id : uuid «FK»
commodity_id : uuid
quantity : int
unit_price : numeric(14,2)
unit_cost : numeric(14,2)
restock_request_items
● id : uuid «PK»
restock_request_id : uuid «FK»
commodity_id : uuid
quantity : int

Sumber: er.puml

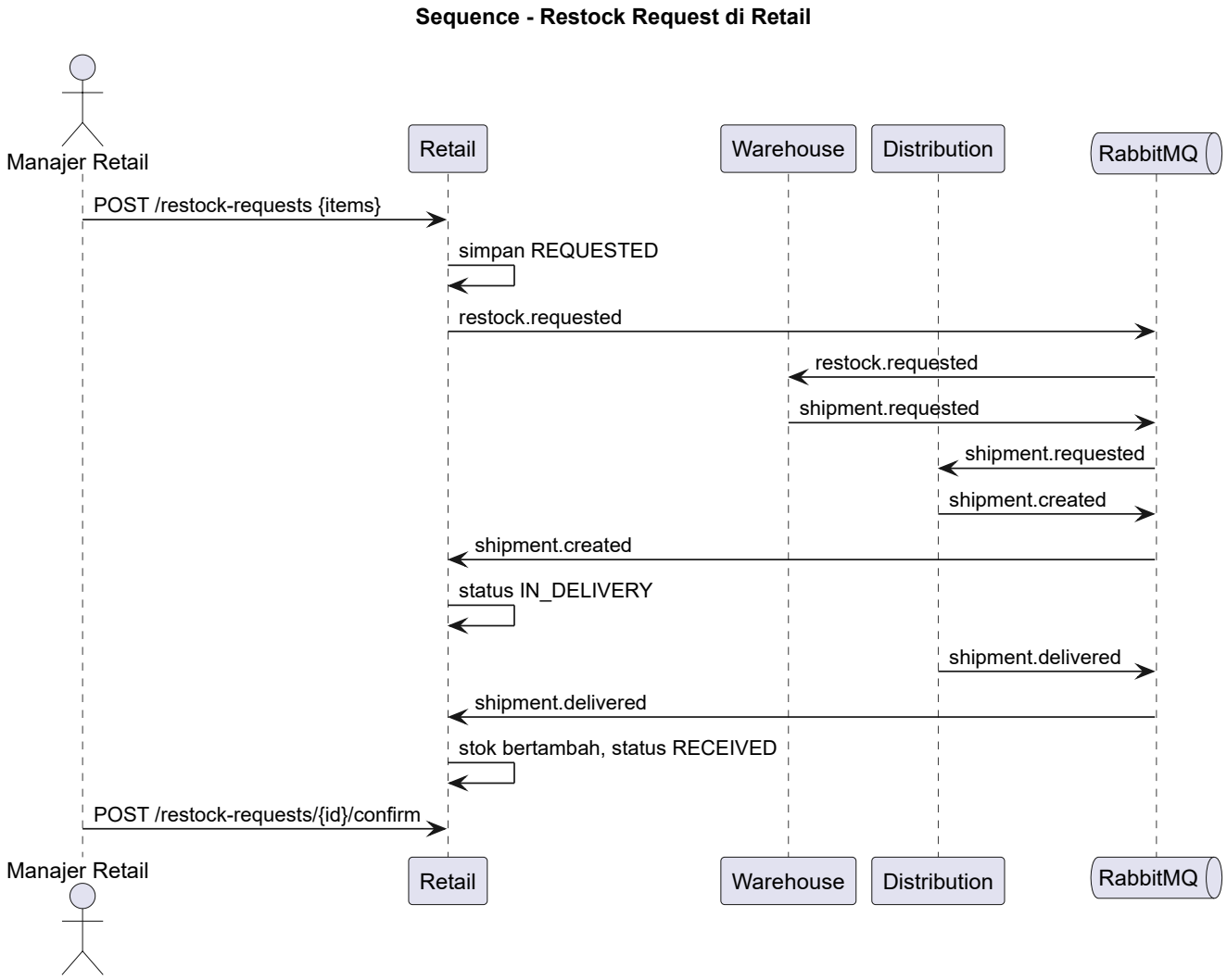
Sequence - Dashboard Retail

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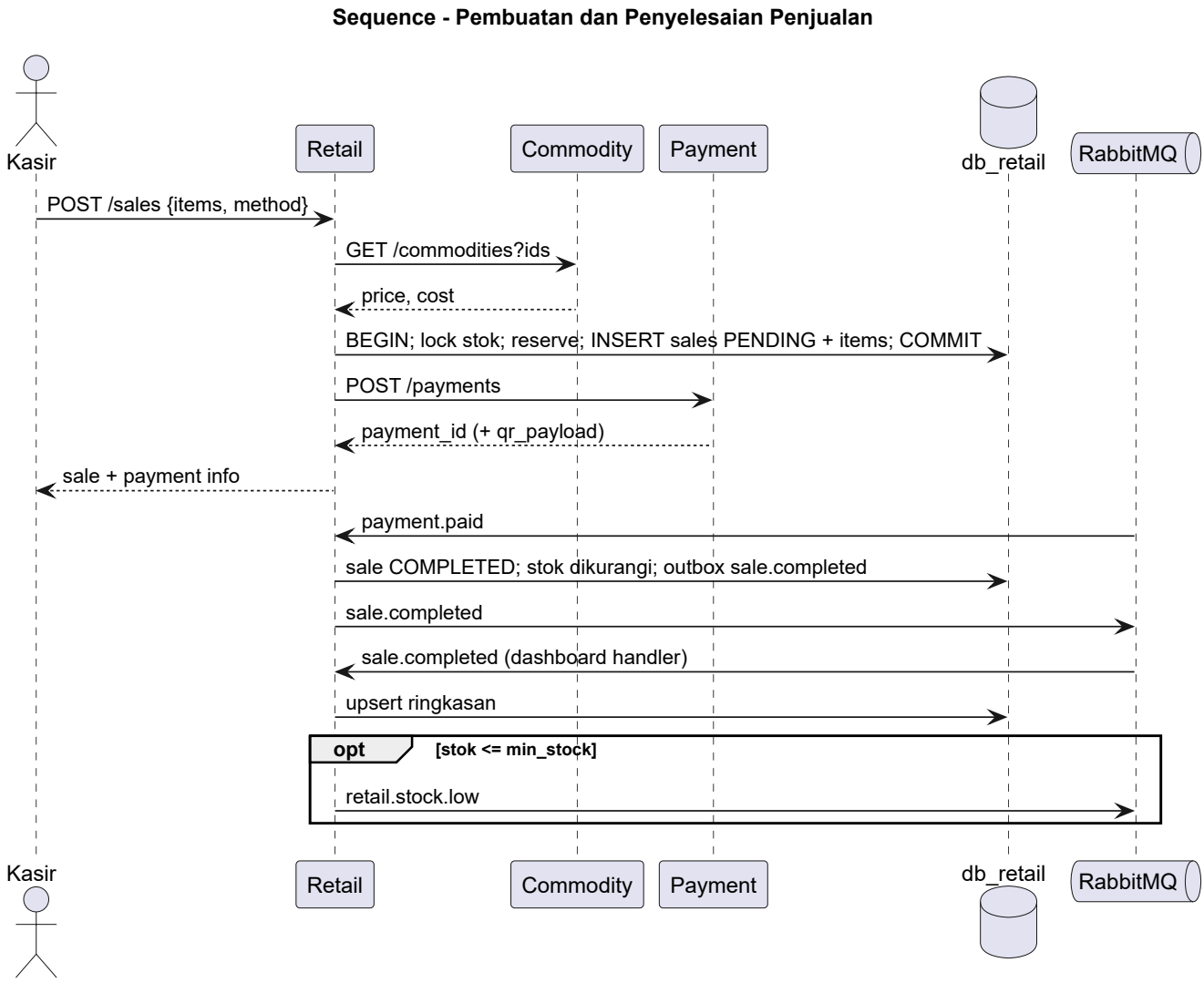
Sumber: sequence_dashboard.puml

Sequence - Restock Request di Retail



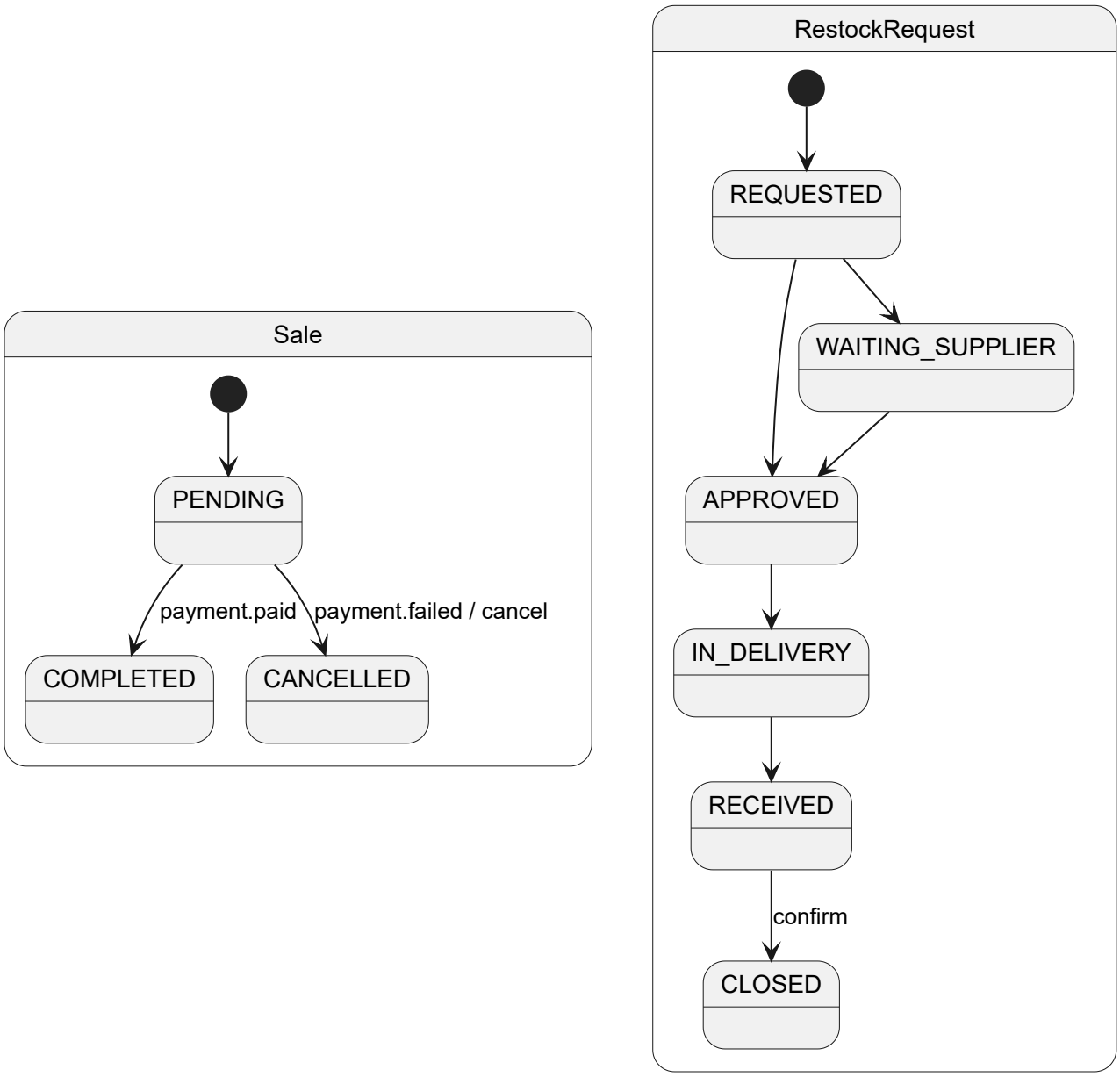
Sumber: sequence_restock.puml

Sequence - Pembuatan dan Penyelesaian Penjualan



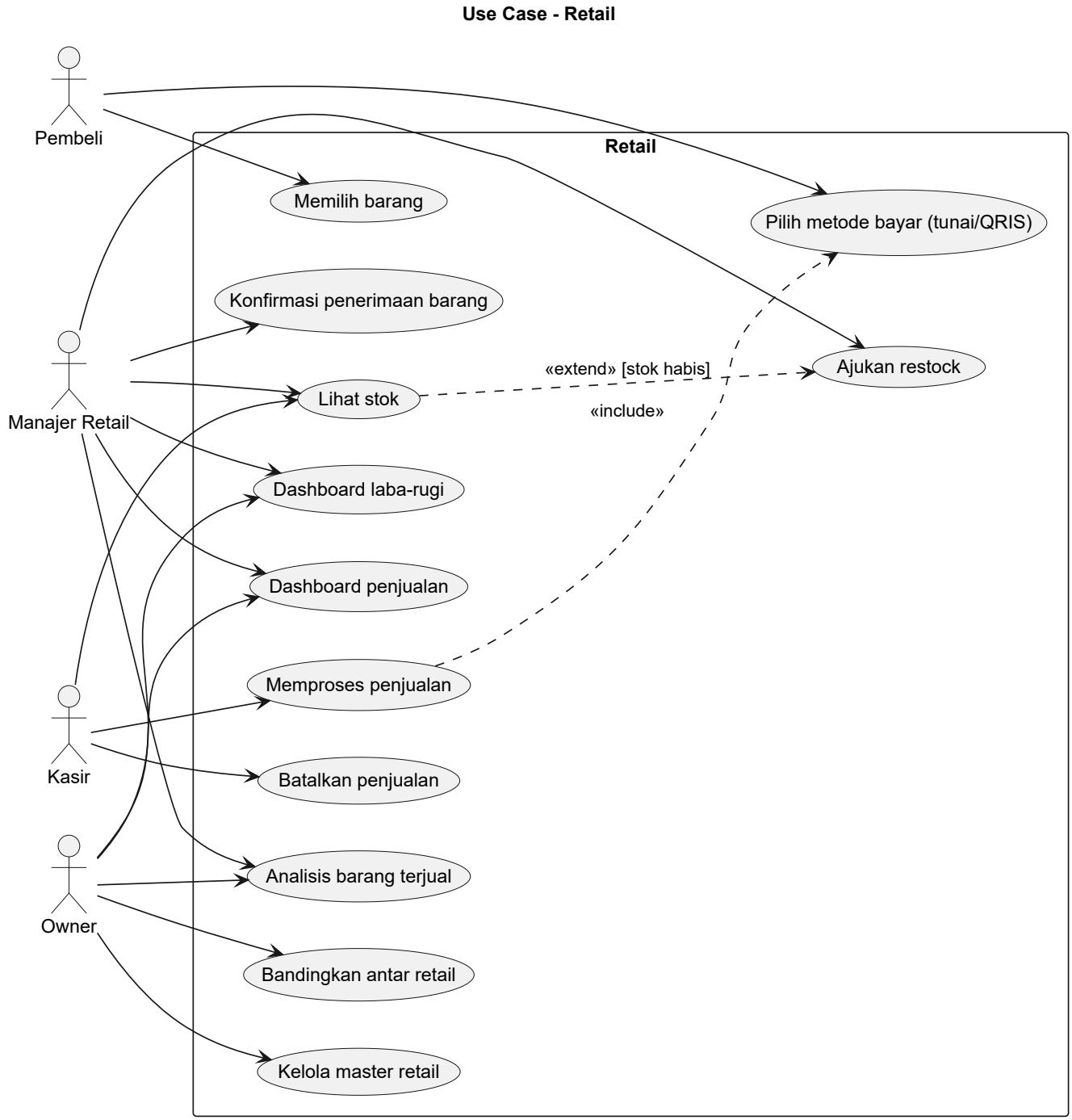
Sumber: [sequence_sale.puml]

State - Sale dan Restock Request



Sumber: state_sale_restock.puml

Use Case - Retail



Sumber: usecase.puml